

4.23 Adjustment of Rewind Layarm Spreader Apex Position

This bow spreader roller has the ability to have its apex position modified by manual interaction.

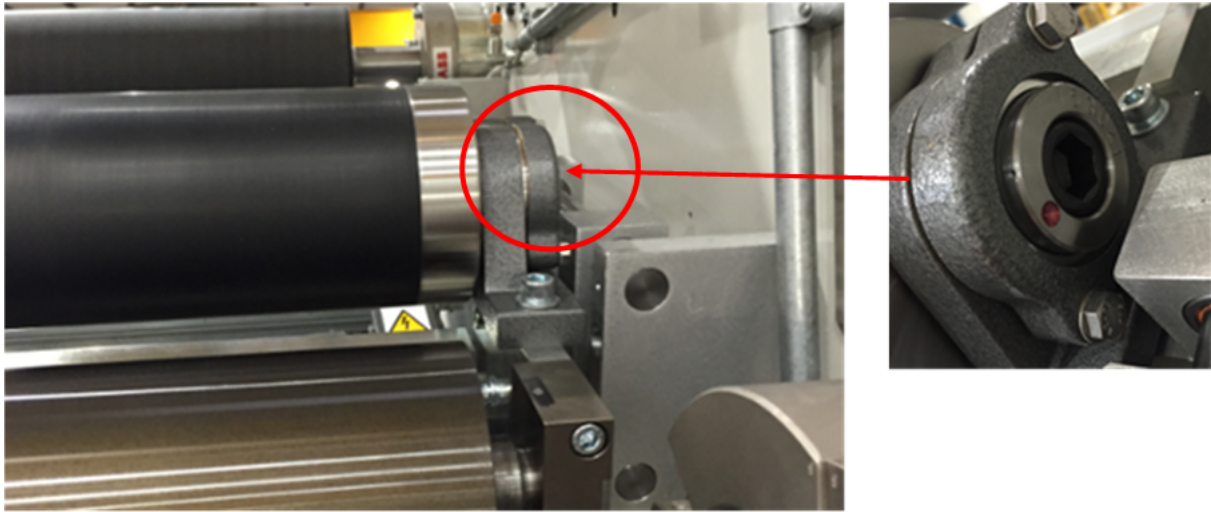


Figure 4.23-1 Apex indicator on the end of the bow spreader.

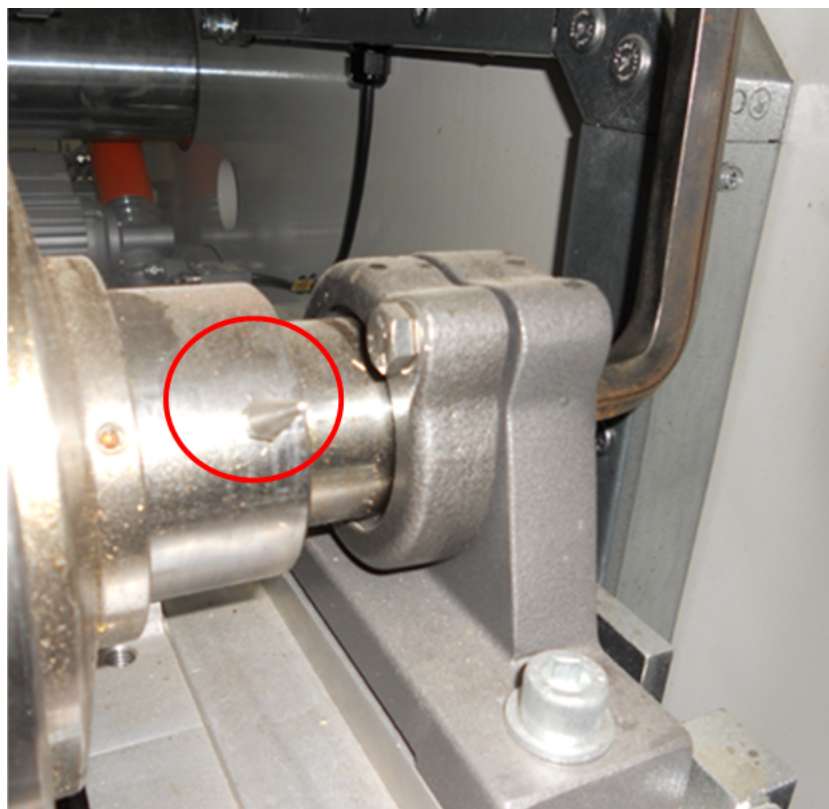


Figure 4.23-2 Apex indicator on the shaft of the bow spreader.

A 17mm Allen key and a 13mm spanner will be required to make the position change.

1. Move the lay-arm assembly towards the rewind core
2. Note the position of the roller bow "apex"
3. Release the two clamping bolts at either end of the roller.

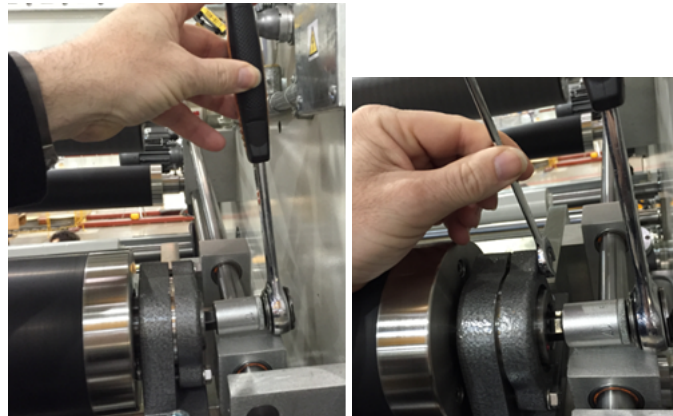


Figure 4.23-3 Examples of a technician releasing the clamping bolts.

Use the 17mm Allen key to move the apex to the required position, and then tighten the clamping bolts.

Remove tools from area.

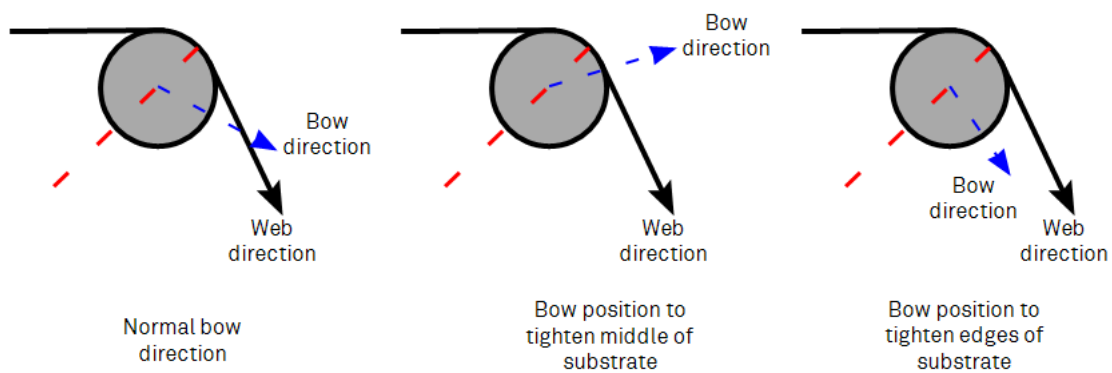


Figure 4.23-4 Bow apex position effects of the spreader roller.

4.24 Unwind Layarm (Gap) Roller

To stop the substrate sliding, feathering or telescoping at the unwind position, a layon assembly is provided.

The unwind layon roller assembly consists of a roller that slides upwards to the unwind roll on the slide assembly.

The principle of operation and control is the same as the rewind layarm.

The major difference is that this assembly consists of a single roller, which is pivoted to allow for differences in the roll circumference.

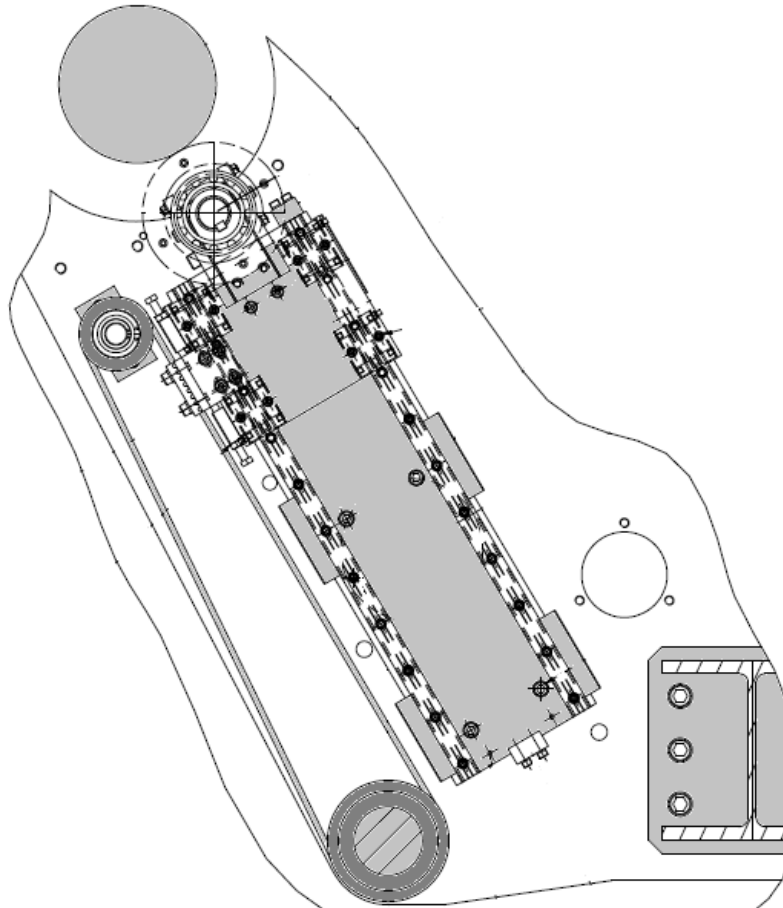


Figure 4.24-1 Unwind layarm pivot assembly roller alignment.

In the "GAP" mode of operation, a small gap (<50mm) is maintained as the roll increases in diameter.

Alternatively, the system can be operated in the "CONTACT" mode, where the last roller before the rewind roll presses on it in contact mode.

	Note !
	If excess pressure is applied, the substrate can be damaged.

Locking the Pivot Movement

In some situations, the pivot action may not be required, for example if it is directing the substrate.

	Note !
	The pivot action can be removed and the arm be locked parallel to the roll. Reference should be made to the assembly drawing.

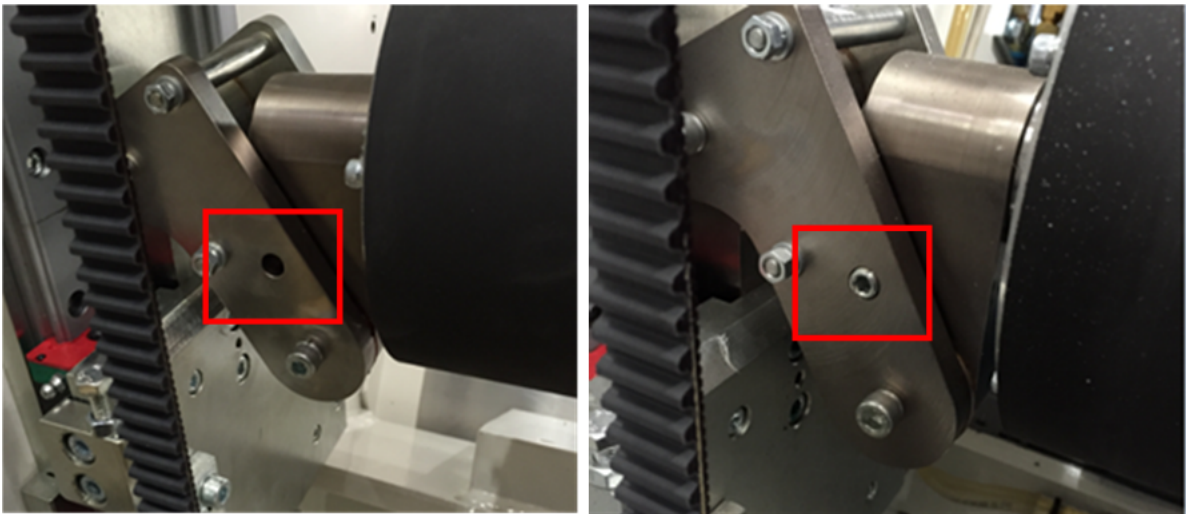


Figure 4.24-2 Pivot assembly with and without locking pin.

Setting the Operating Mode

The mode of operation is set in the RECIPES HMI screen.

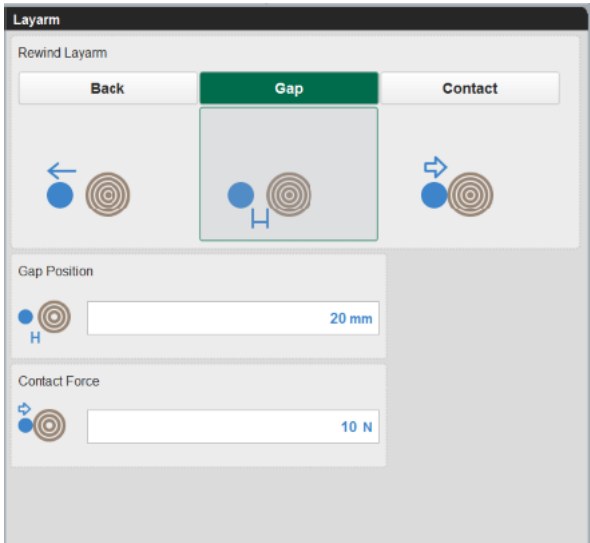


Figure 4.24-3 LAYARM section of the DRIVES RECIPES WINDING HMI screen.

Touching the “Layarm Control Mode” selection button changes the mode of operation.

The amount of contact pressure applied or gap required is set on the CONFIGURATION – WINDING HMI screen. These values can be changed while the coating cycle takes place, but it is advisable not to change between the modes of operation.

	Note !
	Maximum gap that can be set is 50mm, but it is normal to run at 15-20mm. Maximum contact pressure that can be set is 400 Newtons. It is normal to run at 50-100 Newtons.

Movement by Operator

With the chamber open, the layarm assembly can be adjusted by the operator from the rear control station with the "In" and "Out" buttons.

This function is only available with the chamber fully open and winding cart fully retracted.

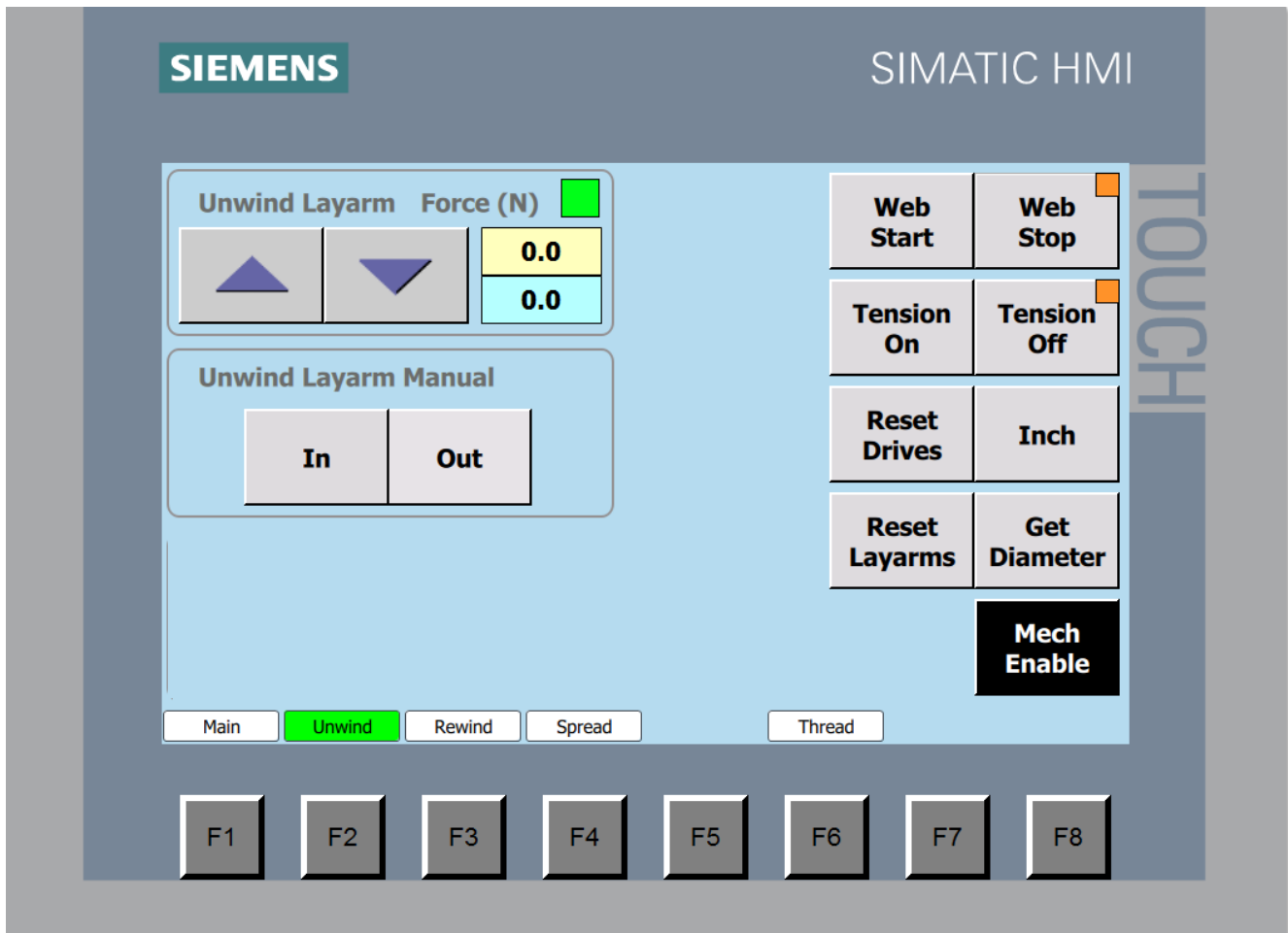


Figure 4.24-4 Rear HMI control station at the loading area.

Limitation of Movement

The upper and lower limits of movement are stopped by mechanical stops at either end of the slide rail.

A proximity switch/sensor also monitors the fully "OUT" (Home) position.

Modes of Layarm Motors during various periods in operating cycle

At certain times in the operating cycle of the machine, the drive system will override the operator's input in the control of the layarms. These situations are described below; -

- Winding cart open, "Reset Layarms" Command
The layarm will return to the home position, when the home switch is enabled. A message on the drive screen will indicate this condition
- Winding cart open, "Get Diameter" Command
The layarm will move from the "home position" to contact the roll/core and stop further movement. A message on the drive screen will indicate this condition
- Winding cart open, tension applied.
The arm will move from if at the "home" position to the roll/core as part of the tensioning up sequence. It will go to the selected mode of operation, either gap or contact.
- Winding cart open, tension applied, inching of drive system
The layarm will track the roll/core in the mode selected.

- Winding cart inside the chamber, tension applied and the drive interlock activated
The layarm will move to the rewind roll. A timer allows the roller to be driven down. It will then apply pressure to the roll of substrate to stop possible movement of the upper layers.
- Winding cart inside the chamber, tension applied, the drive interlock activated and the line is started
When the line start signal is received, the mode of operation should change (as required) to that selected by the user in either contact or gap mode.
- Winding cart moving out of chamber, tension signal deactivated, drive interlock not activated
As the tension mode is deselected, the layarm assembly will be driven back to its "Home position". The limit of this travel is stopped by "home" switch inside the chamber.
- Web break situation with winding cart in the chamber.
Layarm will drive into roll at maximum torque level to minimise possible loss of substrate

4.25 Drum Cleaner Tool

In the situation where the process drum becomes coated with evaporated aluminium, for example when a web break occurs, it has to be cleaned.

The reason for cleaning is to reduce the possibility of damage to the web and to ensure that correct cooling of the substrate takes place.

A cleaning device has been built into the machine to assist the user in cleaning the drum.



Figure 4.25-1 Drum cleaning head in its parked position

Assembly of Drum Cleaner:

The extension handle has to be attached to the section which is left permanently in the machine.

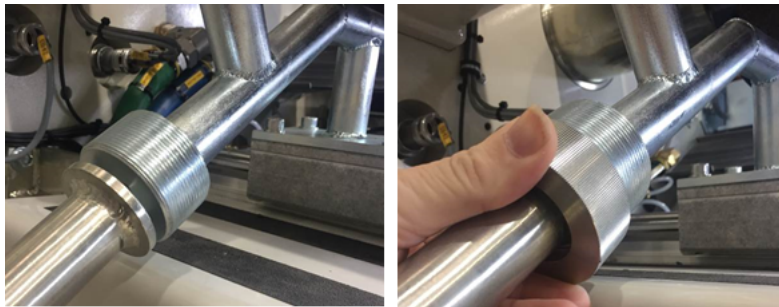


Figure 4.25-2 Drum cleaner extension being fitted by the user

Rotation of the Process Drum

The drive system has been designed to allow rotation of the process drum in a controlled manner by the machine operator. To facilitate this, two devices have been supplied.

- A foot switch is supplied, with a cable that is plugged into a socket on the winding cart panel (E0_6)
- A hand enabled switch, with a cable that is plugged into a socket on the winding cart panel (E0_6)

	Warning !
	The footswitch or hand switch should never be overridden



Figure 4.25-3 Example of an operator using the foot switch



Figure 4.25-4 The hand switch

Direction of Drum Movement

The drum will operate at different speeds and directions dependent upon device used.

- Footswitch: Drum rotates in normal direction at a speed of 15 metres/min
- Hand switch: Drum rotates in opposite direction to normal at 50 metres/min

	Note !
	There are various machine interlocks to stop the machine closing if the activation switch is not removed from the socket.

Use of Vacuum Cleaner

The drum cleaner has been designed to allow the use of a vacuum cleaner to be attached to it to remove any debris as cleaning takes place.



Figure 4.25-5 Example of a vacuum cleaner attached to the drum cleaner



Figure 4.25-6 Example of an operator using the drum cleaner device

Changing Abrasive Pad

To replace the cleaning pad, the design allows from the pad holder to be removed.



Figure 4.25-7 Cleaning head assembly

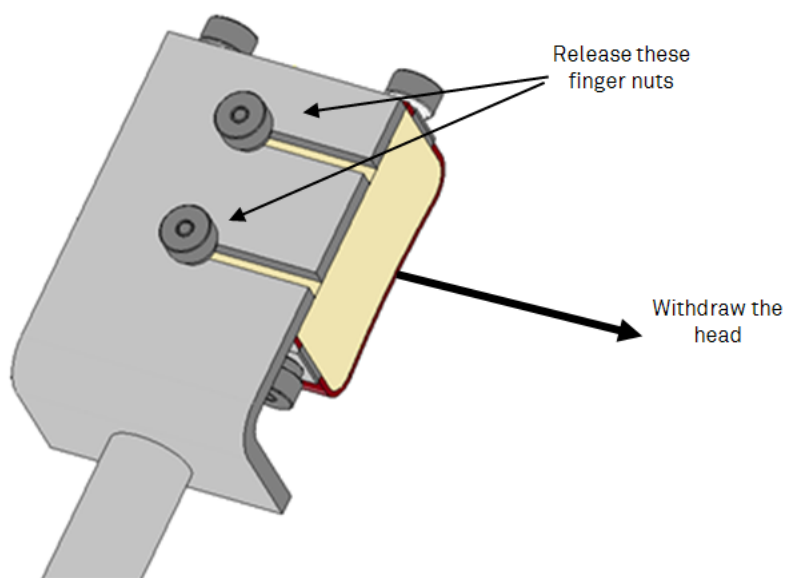


Figure 4.25-8 Cleaning head assembly release bolts

The clamping plates are then removed to allow replacement cleaning material to be installed.

4.26 Unwind Sidelay & Oscillation Functions

This machine has been designed to allow limited side movement of the unwind chuck by approximately 15mm either side of centre, dependent upon the machine specification.

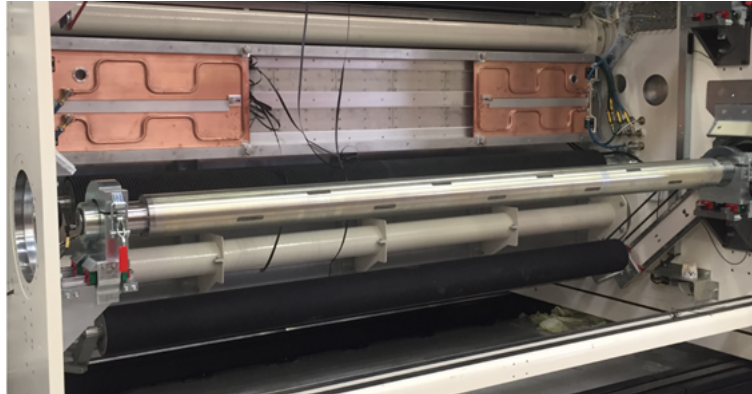


Figure 4.26-1 Complete unwind chuck assembly.

The non-driven chuck is designed to move on a push/pull principle using the reel shaft, to connect the two sections of the chucking system.

Movement is limited by limit switches built into the screw assembly.

The limit of movement is controlled by limit switches located on the outside of the machine as shown in the photograph.

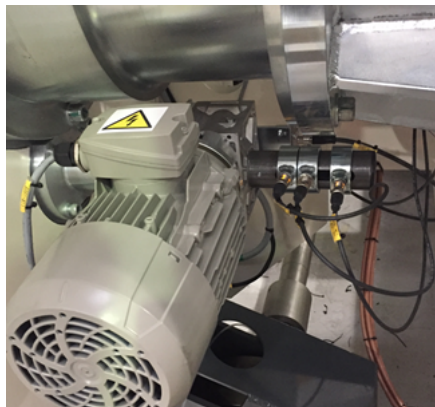


Figure 4.26-2 Unwind sidelay movement motor and switches.



Caution !

Do not override the movement limit switches as damage could occur to the mechanical components.



Figure 4.26-3 Non-driven end of the unwind chucking system.



Figure 4.26-4 Sidelay drive shaft that moves the driven chuck.

Operator Interfacing

The movement can be controlled either from the rear HMI interface unit at the loading area or the main machine HMI at the front of the machine.



Figure 4.26-5 DRIVES HMI Screen indicating sidelay operation buttons.

Green indicator will change to red when full movement in one direction is reached.

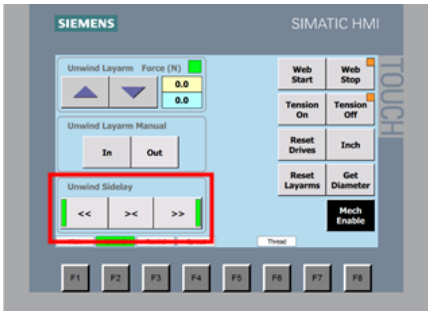


Figure 4.26-6 Control HMI station screen capture - Unwind Layarm.

	Note !
	The chucks have to be closed, and clamped down with a reel shaft in location to use this feature.
	On machines where automatic crane interfacing is included, the system requires the sidelay system to be centralised.

Oscillation Function

The oscillation function is an automated feature that moves the unwind roll by a few millimeters, in either direction and to stop (Dwell) between the movement.

Use the buttons in the screenshots below to start oscillation.

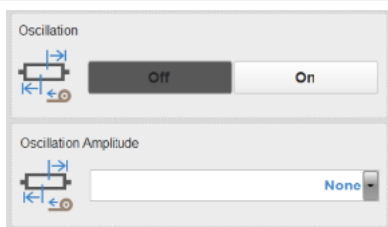


Figure 4.26-7 Oscillation section of DRIVES screen.

The above section of the screen will allow the oscillation (movement) to be set between +/-1mm to 10 mm.

The dwell time is set at 10 seconds, before movement starts.

The oscillation will only take place when the winding cart is in the chamber and the web is running.

	Note !
	The “Zero” point of movement will be from where the roll is located. If it has been moved using the sidelay function, it will be offset.

4.27 Web Threader

A motorised web threader has been supplied with this machine to allow the threading of a “leader tape” through the machine.

The leader tape should be attached to the substrate, to allow the substrate to be manually pulled through the winding mechanism.

	DANGER ! Trap/entanglement risk – Check the area is clear before operating this equipment
---	---

An endless “T-belt” is supported through the winding mechanism on pulleys.



Figure 4.27-1 Web threader belt at the unwind/rewind position



Figure 4.27-2 A leader tape fastened onto the web threader

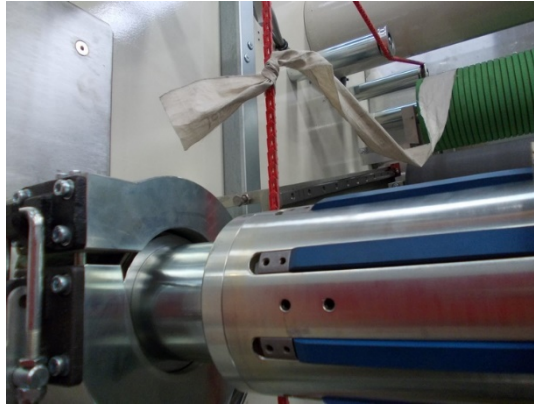


Figure 4.27-3 A leader tape fastened onto the web threader as it approaches the rewind shaft

Operating the threader rotation

	Note !
	The threading tape should be tied to the belt next to the unwind with the knot facing inwards.

Procedure

1. Isolate the movement isolator on the front of the machine
2. Tie a length of threading tape long enough to run through the mechanism to the "red" belt at the unwind position.
3. Turn ON the movement isolator and reset the G-stop circuit
4. With the guard-stop active, movement should be possible from the rear HMI pushbutton, if all the interlocks are correct.



Figure 4.27-4 Example of an operator selecting the threader run button

5. Press & hold the "Run" button from the "Threader" page on the rear HMI.
6. The audible and visible siren will be activated, while there is movement.
7. The threading tape should be pulled through the winding mechanism by the belt and emerge at the rewind position.
8. Release the button, isolate the movement isolator and untie the threading tape.
9. The threader tape can now be attached to the substrate at the unwind and used to pull the substrate through the machine.

4.28 Rewind Sidelay Function & Oscillation

This machine has been designed to allow limited side movement of the rewind chuck by approximately 15mm either side of centre, dependent upon the machine specification.



Figure 4.28-1 Rewind chuck assembly

The non-driven chuck is designed to move on a push/pull principle using the reel shaft, to connect the two sections of the chucking system.

Movement is limited by limit switches built into the screw assembly.

The limit of movement is controlled by limit switches located on the outside of the machine as shown in the photograph.

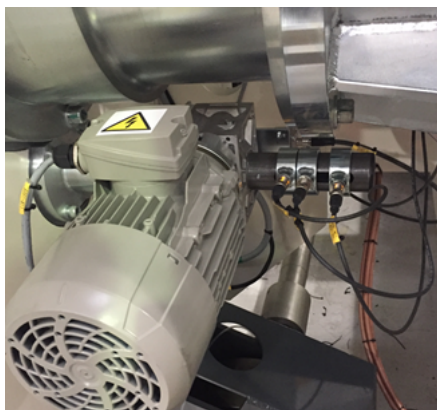


Figure 4.28-2 Rewind sidelay movement motor and switches



Caution !

Do not override the movement limit switches as damage could occur to the mechanical components.

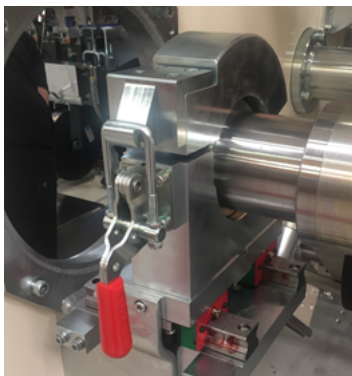


Figure 4.28-3 Non-driven end of the rewind chucking system

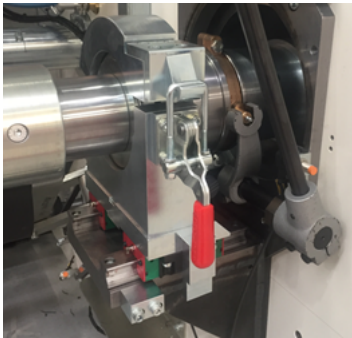


Figure 4.28-4 Driven end of the rewind chucking system



Figure 4.28-5 The sidelay drive shaft that moves the driven chuck

Operator Interfacing

The movement can be controlled either from the rear HMI interface unit at the loading area or the main machine HMI at the front of the machine.

Using either the Sidelay buttons on the main operator HMI on the drive screen, to move the rewind roll within the movement limits.

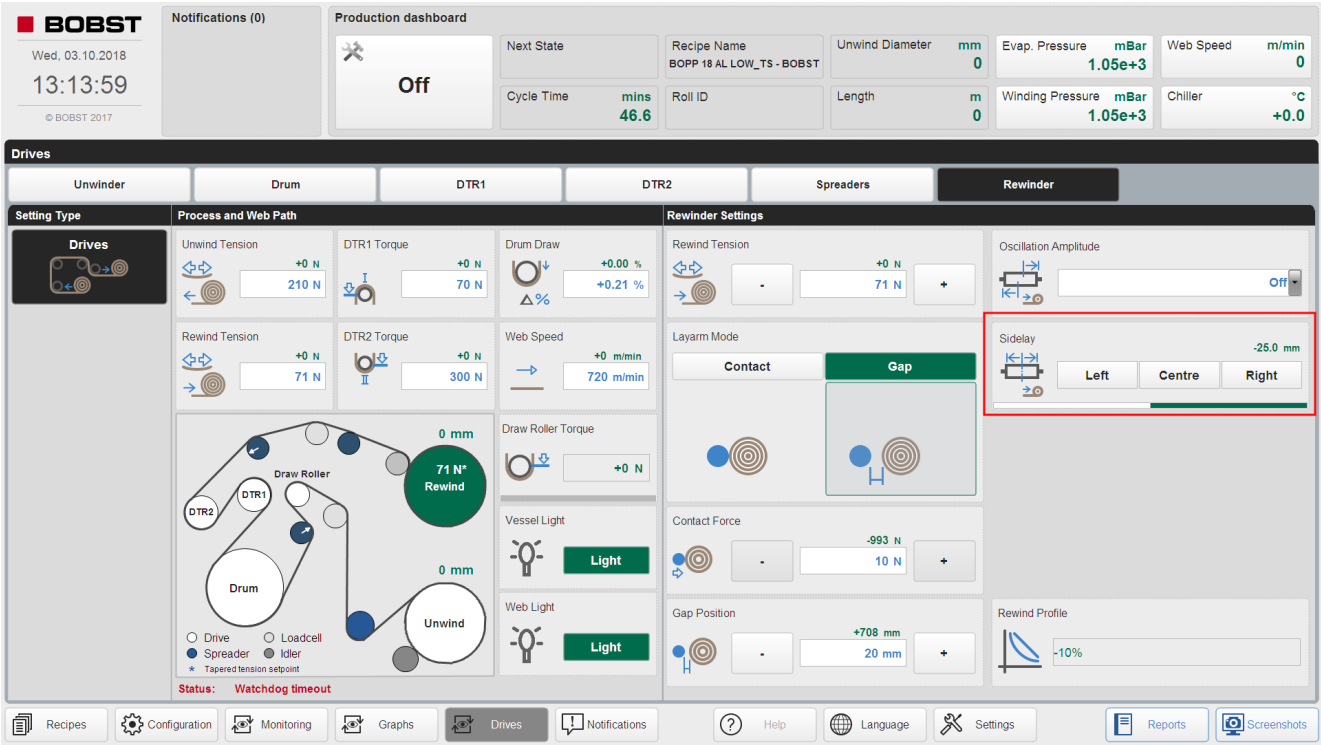


Figure 4.28-6 Drive HMI Screen indicating sidelay operation buttons

Green indicator will change to red when full movement in one direction is reached.

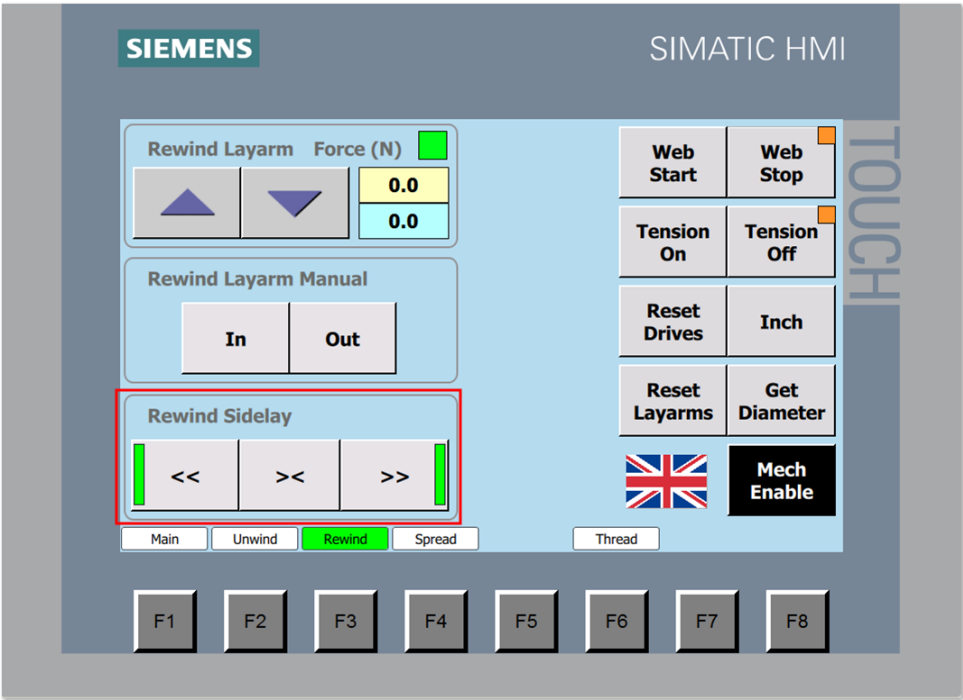


Figure 4.28-7 Control HMI station screen capture - Rewind Layarm

These buttons are used to move the rewind roll left and right, once it has been chucked. Green indicator will go red when full movement is reached.

	Note !
	The chucks have to be closed, and clamped down with a reel shaft in location to use this machine feature. On machines where automatic crane interfacing is included, the system requires the sidelay system to be centralised.

Oscillation Function

The oscillation function is an feature that can be switched on or off. When on, it automatically moves the rewind roll by a few millimeters, in either direction and to stop (Dwell) between the movement.

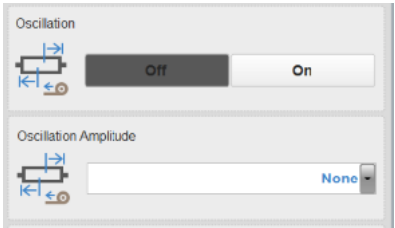


Figure 4.28-8 Oscillation section of DRIVES screen

The above section of the screen will allow the oscillation (movement) to be set between +/-1mm to 5mm
It also allows the dwell time to be set between 0-60 seconds.

The oscillation will only take place when the winding cart is in the chamber and the web is running.

	Note !
	The “Zero” point of movement will be from where the roll is located. If it has been moved using the sidelay function, it will be offset.

4.29 Automatic Crane Interface

A provision has been incorporated into the machine to allow the use of a Bonfanti crane system



Figure 4.29-1 Example of a loading station for the machine



Figure 4.29-2 Example of a crane system loading a roll



DANGER !

Do not override any safety interlock systems installed on either the metallizer or the crane system

Operator Interfacing

The operator interfaces with the machine and prepares the metallizer to allow a full unload/and load sequence from the crane system from the rear HMI.

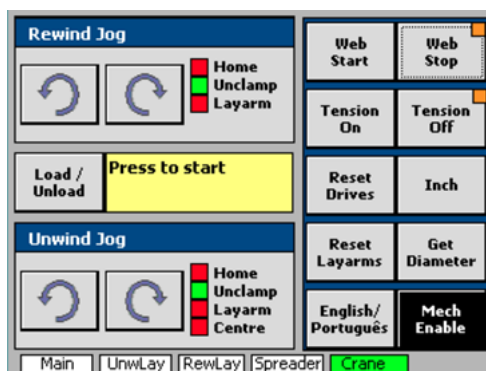


Figure 4.29-3 Dual language local rear HMI interface screen

There is a live message within the yellow background and other status indicators

The crane screen is selected and there are feedback messages, which will appear as the operator prepares the loading area.

Message	Description of Meaning of Message
0 Press to start	Operator presses the Load/Unload button to start the process
1 Check Estop, Gstop and machine open	The machine will check that the following conditions exist Winding cart is fully retracted E-stop safety circuit is enabled G-stop safety circuit is enabled If no issues, the next message will be displayed. If there are issues, check that neither of the reset buttons on the main operator panel are flashing.
2 Check all drives OK and no fast stop	The machine will check if the winding cart drives are healthy and no faults exist. If ok, the next message will be displayed. If the system does not move to the next message, refer to the main machine HMI drive diagnostic screen, and clear any errors.
3 Unwind layarm home diameter measured	The machine will check that the lay arm is at its home position as detected by limit switches at the layarm sides. The message will be still displayed, if the condition is not met.
4 Rewind layarm home diameter measured	The machine will check that the lay arm is at its home position as detected by limit switches at the layarm sides. The message will be still displayed, if the condition is not met.
5 Sidelay centred	Not Applicable to this machine
6 Finding home	This message is displayed when the unwind and rewind chuck are moving to the position where vertical lifts can take place. A sensor detects this and then a local brake will engage to hold this position.

Table 4.29-1 HMI Messages

Message	Description of Meaning of Message
7 Unlock clamps	This is a message to make the operator aware that they need to open the manual clamps to allow the crane to remove/insert the rolls cleanly.
10 Press to call crane	Message to the operator to press the Load/Unload button to start the crane movement to either load, or unload the rolls.
11 Waiting for crane	Status indication that the metallizer is waiting on the crane operation.
12 Waiting for crane	Status indication that the metallizer is waiting on the crane operation.
20 Reset	Indicates that the sequence has been reset.

Table 4.29-1 HMI Messages



Figure 4.29-4 HMI interface screen – Rewind Jog

Manual Jog function

There is a manual function to allow the driven chuck to be rotated, by the operator slowly either clockwise or counter-clockwise. The “home indicator will change colour from red to green, when the chuck is in the correct position.

Indicators

An explanation of the indicators in this area of the screen are as follows.

Indicator	Description of Meaning of Message
Home	Chuck is in the correct position to allow removal or placement of a roll/shaft
Unclamp	Indicates that the chuck is mechanically unclamped by the operator
Centre	Indicates that on machines with sidelay function, the chuck has been centred in the machine.
Layarm	Indicates that the layarm has been positioned at its home position.

Figure 4.29-5 HMI Messages

Resetting sequence to “Reset” condition

If the operator needs to reset the sequence, this can be done by pressing the load / unload button on the HMI for 5 seconds.

Diagnostics Screen

In the case where the machine will not allow the crane system to operate, because a condition/interlock not being met, a diagnostic screen is supplied.

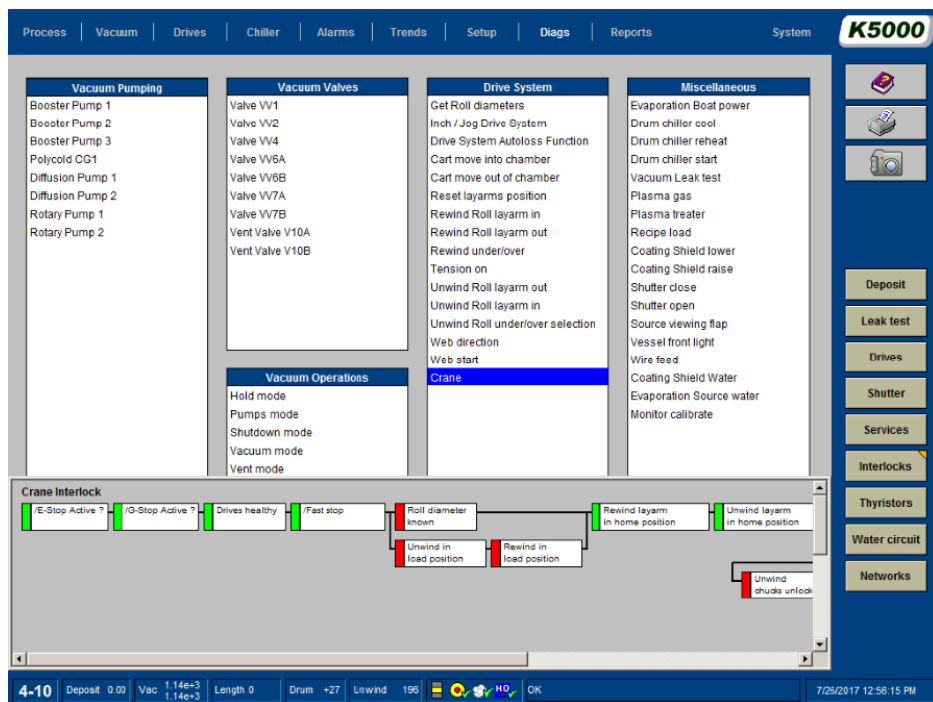


Figure 4.29-6 HMI diagnostic screen - Crane

The following indicators are shown, the table below explains what they relate to. For successful operation all should be green, except where there are two legs in parallel, when one leg should be green.

Indicator	Description
E-Stop Not Active	Machine emergency stop has not been tripped
G Stop Not Active	Machine Guard circuit/isolator has not been tripped
Drives Healthy	All the winding mechanism drive components have to be healthy
Not fast stop	Drive system not registering a Fast stop, caused by a web break situation.
Roll Diameter Known	The drive system has gone through a “Get diameter” procedure, and the roll diameters are known.
Unwind in load position	The sensor at the unwind chucks register that there is a shaft in location
Rewind in load position	The sensor at the rewind chucks register that there is a shaft in location
Rewind layarm in home position	The rewind layarm is at its home position away from the roll/core
Unwind layarm in home position	The unwind layarm is at its home position away from the roll/core
Unwind sidelay centred	If there is a sidelay function on the machine, the machine PLC checks that the shaft is back to the central position,
Unwind in Load position	The sensor at the unwind chucks register that there is a shaft in location
Rewind in load position	The sensor at the rewind chucks register that there is a shaft in location

Table 4.29-2 HMI Diagnostic Conditions

Indicator	Description
Unwind Chucks Unlocked	Indicated when the mechanical clamp is released and the spring pressure lifts the bearing clamp
Rewind Chucks Unlocked	Indicated when the mechanical clamp is released and the spring pressure lifts the bearing clamp
Winding Cart fully open?	Indicates when the winding cart is fully retracted from the chamber
Unwind Brake fail	Air pressure has to be monitored being applied to the holding brake and the position sensor is activated
Rewind Brake Fail	Air pressure has to be monitored being applied to the holding brake and the position sensor is activated

Table 4.29-2 HMI Diagnostic Conditions

SECTION 5.0 ALUBOND PROCESS

The Alubond process option, including hardware is supplied with this machine. This section of the manual describes the Alubond process in the following chapters:

- [5.1 Overview of the Process on page 241](#)
- [5.2 Safety Considerations on page 243](#)
- [5.3 Hardware on page 259](#)
- [5.4 Machine Set Up on page 267](#)
- [5.5 Process Guidance - Conditions on page 271](#)
- [5.6 Cleaning Methods on page 275](#)
- [5.7 Testing of Processed Material - Coating Adhesion on page 281](#)
- [5.8 Process Start-Up/Operation/Shutdown Check Sheet on page 283](#)

5.1 Overview of the Process

Under standard aluminium metallizing conditions, a critical process parameter, (post metallizing) is the ability of the aluminium to adhere to the surface of the substrate. This is commonly known as the “adhesion”.

Historically, special expensive base substrates (chemically treated) have been produced to give “high adhesion” between the aluminium and the substrate.

Bobst Manchester has developed the ALUBOND™ process to provide consistent high level of aluminium bond strength (adhesion) in a production environment on low cost substrates.



Figure 5.1-1 Example of the finished product

This is independent of the type of pre-treatment process.

Outcomes of poor metal adhesion

It was found that poor adhesion led to the transfer of the aluminium between metallized substrate and the adhesive in a typical laminate structure.

Traditional metallization (even with Plasma treatment) can result in poor metal to polymer substrate bonding, this produces de-lamination which leads to the package failing

Metal adhesion has long been a challenge in the converter packaging world due to this inter-layer bonding failure.

Bobst recognised that there was an increasing demand globally for higher levels of metal adhesion on metallized film due to the industry's requirement for more complex packaging structures which requires a lamination peel strength suitable for functional needs.

Part of the problem is low surface energy on the metal side which also contributes to poor lamination bonding;

De-lamination leads to product rejects and loss of reputation.

What does Alubond™ deliver?

Bobst Manchester has developed an innovative approach to address this common issue in the packaging industry through its vacuum metallizers, eliminating the need for chemically treated films.

High bonding strength is achieved on any substrate with the BOBST AluBond® unique metallizing process.

This is an advanced metal adhesion technology where metal adhesion values can be achieved of up to 5N/15mm.

By comparison, normal levels are for example 0.5N/15mm for PET substrate.

In some cases, adhesion levels are so high that it even leads to film tear during the peel test.

How is the Alubond™ process carried out – The basic principles

A seeding layer is created in the process to enhance the adhesion between the aluminium and the base substrate film.

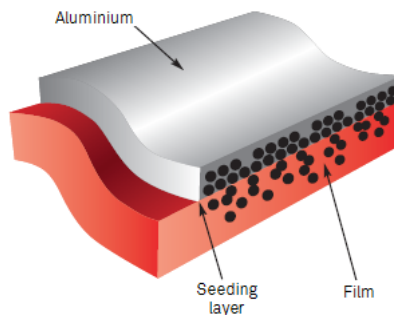


Figure 5.1-2 Seeding layer

1. Aluminium wire is evaporated from the standard evaporator's boats in a modified aluminium vacuum coater.
2. Oxygen gas is injected into the aluminium vapour cloud which reacts to form a seeding layer on the substrate.
3. The amount of oxygen gas is fixed, while the aluminium wire is varied to give a consistent optical density, as measured using the Hawkeye in-line monitor system.
4. This equipment is described briefly in this section of the manual.

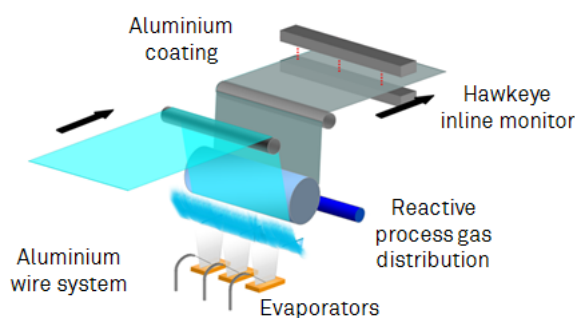


Figure 5.1-3 Process Schematic

Secondary Benefit – Dyne Level Retention

It has been found that a secondary benefit of the process has been shown to significantly increase dyne level retention on the finished product.

This is an added benefit to the improved metal bonding strength which translates in improved ink wettability during printing and enhances structure stability during lamination. BOBST AluBond® generates added value by extending surface energy stability on metallized films over a long period of time and may eliminate the need for an additional surface treatment boost i.e. corona refresh prior to converting.

Dyne level reducing-factors over time play a detrimental role in film composite structures; BOBST AluBond® delays dyne level decay issues and facilitates creation of a robust, fit-for-purpose laminate.

5.2 Safety Considerations

	Note !
	This information is supplied as an addition to the SAFETY part of this manual, as it is specific to the process, being carried out. All other risks that have already been identified in the safety section still apply.

As the machine uses oxygen as a process gas, there are a number of safety features that have been incorporated into the machine to ensure the safe delivery of this supply from the customers supply point to the process area.

	Warning !
	

The machine oxygen delivery system and vacuum pumps exhaust system have been designed to comply with good industrial practices.

At the exhaust from the vacuum pumps, the concentration of the oxygen should be less than 23% by volume.

Do not override any of the safety interlocks built into this machine.

Guidelines have been provided at the time of machine order placement to end customers, in regards to the installation of the machine, exhaust systems for the vacuum pump emissions and PPE requirements.

The coating produced during the Alubond™ process is also potentially a hazard, as the material is not fully reacted in some situations. The waste product is described as non-fully reacted AlOx.

This document will highlight “good practices” that should be adopted from an operation aspect.

	Warning !
	
	

It is the customer's responsibility to assess the asphyxiation risk from process gas leaks at their site, considering the room size and ventilation of room and pit area under the machine.

The management of this waste is also the responsibility of the customer. Guidance is given in this section on precautions and systems which can be employed to minimise risk, as identified by Bobst Manchester Ltd.

A system of regular checking is recommended to ensure all machine safety systems are functioning and that conditions are safe prior to running this process on the machine. An example document is provided.

The machine has been designed in such a way as to reduce these risks as much as possible without obstructing the machine's intended function.

Safety considerations are mentioned in the appropriate sections throughout this manual. Detailed advice is given on risks and recommended safe practices however the ultimate responsibility for safe machine operation lies with the customer.

	Note !
	The process materials listed below are specified for operation of this machine. The vacuum pumping system is designed for the listed specified gases. No other process materials should be evaporated or other process gases be used other than those listed. Use of other materials or gases is considered to be improper.

Specified process materials:

Process materials	Designation	Chemical Symbol
Gases:	Argon	Ar
	Oxygen	O2
	Nitrogen	N

Table 5.2-1 Specified Process Materials

Process Gas Delivery System

The gas delivery system is manufactured from stainless steel pipework and other components rated to the international standards for 100% oxygen gas delivery systems.

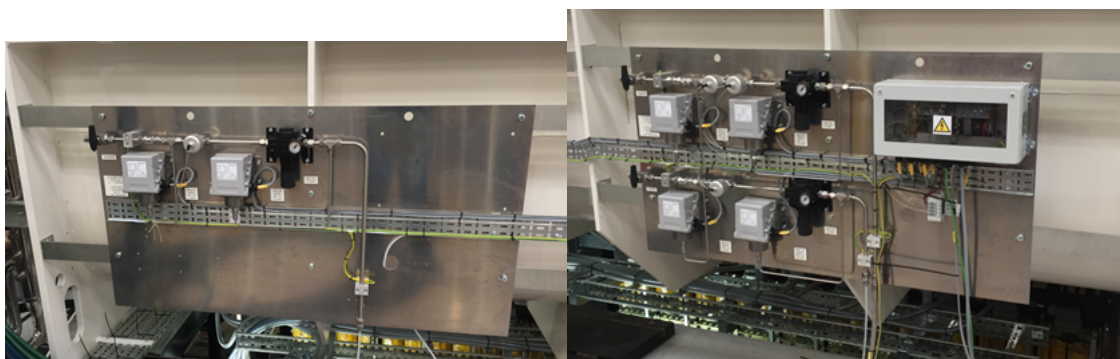


Figure 5.2-1 Components of the process gas distribution panel



Figure 5.2-2 Process gas MFC controllers

	DANGER !
	If any maintenance, component change out, etc. is to be carried out on the process gas delivery system, this should be carried out to the guidance given in BML document 327G821

Machine Venting from Vacuum Condition to Atmosphere

Thin aluminium oxide coatings are classified a ceramic material and therefore does not release any heat contained within the coating as quickly as pure aluminium coatings. Therefore it potentially retains enough heat that could cause a chemical reaction once the vacuum chamber is vented to atmosphere and opened.

When this process is used, the venting sequence changes from a fast vent to a slow vent, automatically. This vent period is programed to ensure that there is a controlled environment, inside the machine. This is provided to

minimumise the possibility of a reaction taking place inside the chamber, and to allow the waste deposit coating on the collection shields to be surrounded by the air gas molecules slowly.

	DANGER !
	Do not modify the venting sequence for the machine. Do not manually open any other valves, etc. to reduce the safe venting method.

Monitoring For Reactions of The Deposit on Collection Shields

There are temperature monitoring thermocouples attached to the collection shields between the evaporation source and the chamber wall, to monitor the temperature of them. If a reaction takes place and there is a fire, these sensors will bring up an alarm and place the machine in a hold condition, stopping further venting.



Figure 5.2-3 Thermocouples on underside of the shield

	DANGER !
	Do not operate the machine for this process without these interlock devices being installed.

Vacuum Pumps Dilution and Exhaust System

To ensure that the concentration of oxygen being exhausted from the vacuum pumps when the Alubond™ process is being used, or oxygen is being supplied to the plasma treater(s). Oxygen is restricted to below 23% by volume, by air being fed into the pump(s).

The air is pulled into the pump, by vacuum half way along the pump body at a regulated value.

There are two flow monitors used to ensure enough air is being drawn into the pump.



Figure 5.2-4 Air dilution air flow monitor system

	Warning ! Do not modify the setting of this interlock system The machine oxygen delivery system and vacuum pumps exhaust system have been designed to comply with good industrial practices. At the exhaust from the vacuum pumps, the concentration of the oxygen should be less than 23% by volume. Do not override any of the safety interlocks built into this machine. Follow the guidance on checking the operation before carrying out the process.
	Note ! The machine is interlocked not to allow the process to operate, if there is any issue with this gas dilution.

A drawing, is supplied to the customer during the project/build stage prior to the machines arrival on site.

- BML 421G193/302 Exhaust Duct Instructions
- BML 421G194/0 O2 Exhaust Duct Instructions
- BML421G192 O2 Exhaust Duct Extraction System User Guide

These are provided to allow the customer to design the duct system based upon the good practice notes contained within the document.



Figure 5.2-5 Examples of customer exhaust systems

Access to the Pit Area

Oxygen has a specific gravity 1.113. In comparison, air is 1.0.

This highlights that oxygen gas is heavier than air and may accumulate in confined spaces, particularly at or below ground level.

Anyone working on the machine in confined spaces needs to understand this fact, and if proposing to enter the pit area under the machine –checks must be made to check that there is no build-up of any gas.

It is advised that an oxygen sensor be installed in the pit area, and using a remote display, the concentration of the oxygen, in the area can then be checked before entering.

<https://www.amio2.com/percent-oxygen-analyzers>



Figure 5.2-6 Example of an oxygen analyser

	Warning !
	Do not enter the area under the machine at a low level without checking the quality of the air in this area

An alternative system uses a remote sensor and display module, placed at the entry of the pit.



Figure 5.2-7 Example of an oxygen deficiency monitor that provides remote information

Forced Ventilation

It is also strongly recommended that “forced” air ventilation is provided to this pit area to reduce the possibility of toxic gas build up.



Figure 5.2-8 Example of forced air circulation pipework in a pit, under the main machine chamber.

	Warning ! Bobst Manchester recommends that a second person at floor level is required to oversee the entry into the area. They should also be in frequent communication with the person entering the enclosed area.
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	Note ! Many companies operate a “Enclosed Space Entry” restriction system and a “permit to work” policy
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A source of recommendations of good practice can be located at:-

<http://www.hse.gov.uk/pubns/indg258.pdf>

As previously commented, the AlOx coating is also potentially a hazard, as the material is not fully reacted in some situations. The following equipment should be provided to minimise the potential hazard.

Fire Extinguishing Equipment

Due to the possibility of the coating not being fully reacted, there is a higher possibility of a fire when cleaning takes place of the machine. To extinguish the fire, it is recommended that ONLY a type/class D extinguisher be used.

Class D fires are described as

“Fires caused by burning metals that combust easily on contact with air, such as magnesium and lithium”

Such specialised fires require a specialised metal powder fire extinguisher to deal with them, and will be required where manufacturing processes involve the risk of metal fires, such as in this process.

It is recommended that these extinguishers are fitted with special low velocity applicator lance, which prevents the spreading of the burning debris, as it slows down the delivery of powder to allow the user to carefully aim where it is needed.



Figure 5.2-9 Class D extinguisher

	Warning !
	Do not attempt to extinguish a fire with any other type of extinguisher, as they could make the situation more dangerous DO NOT use CO2 or Water extinguishers



Figure 5.2-10 Example of Alubond/AlOx waste storage equipment and fire extinguishers

	Note !
	As shown, all the storage equipment is located on top of a containment pallet, which is filled with “Dry Kiln Sand”. This is a secondary safety device, in the case that a fire occurs, and the base of the drum “burns through”

Use of Dry Kiln sand

Dry kiln sand can be used to suppress a waste debris metal fire together with suitable storage equipment.

	Warning !
	If the sand is damp, it will potentially act as a reactant to the non-reacted debris

If a fire ignites in one of the storage containers, it is advisable after it has been extinguished that this sand is placed over the debris to “smother” the material and reduce the possibility of it igniting again



Figure 5.2-11 Example of a kiln dried sand storage container with the lid removed, local to the debris container.

	Warning !
	It is suggested that the drums are stored as shown

Use of a fire blanket

It is recommended that a fire blanket be kept in the locality of the machine, in the case of a fire occurring and there is a requirement to use it for “smothering” purposes.

	Note !
	The fire blanket should comply with “EN 1869:1997 - Fire Blankets” to be used on metal fires

Storage of waste AlOx process materials

All waste materials (storage containers, vacuum cleaners etc.) used in the factory should be clearly marked and left in a clearly marked area away from any potential source of ignition or materials that can burn in the event of a Al metal fire (i.e. factory walls)

Internal and external storage areas can be equipped with smoke detectors and be built from flame resistant materials.



Figure 5.2-12 Example of an internal storage area for the waste material

Note that the storage containers are located on a cart, which has a permanently attached moving device to allow for it to be removed from the internal location to outside quickly.

	Warning !
	Ensure that there is a marked clear path from the internal location to an external storage area.

All storage bins used at the machine, for AlOx debris, should only be partly filled (2/3) to reduce the fuel source of the fire and also fitted with special fire suppression lids.

Large metal containers (45 gallon) are suitable for storing larger quantities of AlOx debris away from the factory/production area.



Figure 5.2-13 Manufactured suppression lid, as supplied by Bobst.

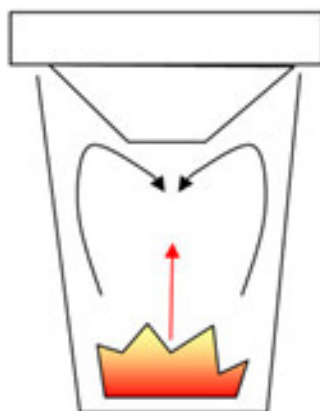


Figure 5.2-14 Image demonstrating the effect of the suppression lid in reducing the air needed for combustion to take place

A cart is provided to be placed under the coating shield area to minimise the contaminated area when cleaning the debris from the shield



Figure 5.2-15 Example of a cleaning cart

	Warning ! Remove all debris at the end of each cleaning cycle from the cleaning cart
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A smaller container is used at the machine to place the AlOx debris in.



Figure 5.2-16 Example of a small debris container

	Warning ! Empty the container at the end of each cleaning cycle
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Figure 5.2-17 Example of a clearly identified 45 gallon drum